

The Resilience Intelligence Fabric (RIF™) and Constraint-Aware Context Matching (CACM™)

A research-backed knowledge graph approach to converting disruptions into executable recovery and optimisation plans

EcocomityChain.AI · January 2026 · v2.0

Executive Summary

Most Supply Chain Risk Management (SCRM) platforms stop at visibility: they detect risk, raise alerts, and open tickets. ECC is engineered for a different outcome — **faster time-to-action and faster time-to-recovery** through evidence-grounded, constraint-aware recommendations and optimisation-backed execution plans.

What makes this approach distinctive is the combination of two complementary knowledge graphs, connected by an intelligence layer — the Resilience Intelligence Fabric (RIF™) — powered by a proprietary matching algorithm: Constraint-Aware Context Matching (CACM™). Recommendations are packaged into executable recovery plans using mathematical optimisation rather than generic playbooks.

Core principle: Disruption response is a race between two clocks — the clock of operational damage and the clock of organisational coordination. ECC compresses both by providing actionable options with evidence, validated feasibility, and optimised execution — all in one flow.

From SCRM to Supply Chain Intelligence

ECC addresses a different problem from traditional SCRM platforms. The shift is from risk dashboards to intelligence-backed decisions:

Traditional SCRM	ECC Supply Chain Intelligence
Detect — Alert — Ticket — Track	Understand — Decide — Execute — Learn
Visibility and compliance workflows	Action synthesis under constraints + optimisation packaging
Generic best practices	Research-backed incident evidence + measurable levers (ICA/PCA)
Hard to translate alert to action	Built to reduce time-to-action and time-to-recovery

The Two-Graph Architecture

ECC maintains two distinct but complementary graphs. Keeping their roles crisp is essential to understanding how the platform works.

Operations Graph (Customer Digital Twin)

This is the customer's live supply chain reality: product structures, parts, materials, suppliers, plants, lanes, inventory, lead times, and constraints. The Product Material Genealogy (PMG) model sits here. This graph answers two questions: **what is impacted?** and **what is feasible right now?**

- BOM — assemblies, sub-assemblies, parts, raw materials
- Plants, work centres, capacities, and key constraints
- Inventory positions, ATP/shortfalls, allocation rules
- Supplier tiers, lanes, lead times, transit constraints
- Orders and commitments (customer demand / planned production)

Resilience Research Knowledge Graph (Global Playbooks)

This is ECC's research-backed intelligence asset: a structured library of incidents, mitigation measures, outcomes, constraints, and evidence pointers. It is not a digital twin of the customer's operations — it is the playbook engine. This graph answers: **what actions exist?** and **what has worked before?**

- Historical disruptions across multiple industries and genres
- ICA measures (immediate corrective actions) with evidence traces
- PCA strategies (permanent corrective actions) mined over time
- Constraints and applicability envelopes — when a measure works and when it fails
- Extension-ready schema to incorporate new categories (tariffs, supply chain finance, sustainability)

The Operations Graph answers what you can do. The Research Knowledge Graph answers what you should consider. RIF connects the two.

Resilience Intelligence Fabric (RIF™)

RIF is the intelligence layer that binds the two graphs and makes them operational. It turns resilience intelligence into executable decisions.

RIF Responsibilities

- Normalise disruptions into structured scenarios (symptoms, constraints, blast radius)
- Retrieve and rank relevant incidents, measures, and evidence
- Validate feasibility against the Operations Graph (inventory, capacity, lead time)
- Package an optimised plan using mathematical optimisation (allocations, alternates, buffering, prioritisation)
- Drive a guided response workflow: evidence → decision → execution → feedback

CACM™ — Constraint-Aware Context Matching

CACM is the core algorithmic engine inside RIF. Unlike simple similarity search, CACM matches the current scenario to historical disruptions while simultaneously respecting constraints from the Operations Graph (what is feasible today) and from the scenario itself (what must be

preserved).

The CACM pipeline runs in four steps:

1. **Normalise scenario** — symptoms, constraints, blast radius
2. **Retrieve incidents** — cross-industry evidence and outcomes
3. **Match under constraints** — feasible levers only
4. **Package plan** — optimisation-ready recovery plan

CACM produces three classes of output: **action candidates** (ICA measures ranked by evidence relevance), **feasibility filters** (actions that violate constraints are eliminated or modified), and **plan packaging inputs** (structured levers for mathematical optimisation).

CACM is not generic retrieval. It is a constraint-aware measure synthesis engine that bridges evidence and feasibility — recommending actions that worked in comparable incidents, filtered by what is feasible for the customer today.

Reducing Time-to-Action and Time-to-Recovery

Disruption tools often generate awareness faster than organisations can respond. The real bottleneck is the **decision-to-execution gap** — a long cycle of interpretation, debate, feasibility checks, and manual plan creation.

ECC compresses this cycle through five mechanisms:

- **Pre-structured options:** measures arrive as structured levers, not prose
- **Evidence attached:** each measure includes the reasoning and contexts where it worked
- **Feasibility-aware:** measures are filtered against current inventory and capacity constraints
- **Optimisation-backed:** plans are packaged as optimised allocations rather than human guesswork
- **Guided workflows:** a structured flow UX reduces ambiguity and aligns stakeholders faster

Practitioners consistently emphasise that dashboards alone do not shorten recovery time. What they need is actionable strategies — "tell me what to do next, and whether it will work under my constraints."

Keeping Measures Fresh and Continuously Expanding

A major weakness in static playbooks is staleness: disruptions evolve, regulations change, and new mitigation patterns emerge. ECC is designed so that new knowledge becomes available to customers without re-platforming.

Why the Research Knowledge Graph Stays Current

- Continuous ingestion of public research, case studies, and incident reports
- Structured extraction into consistent taxonomies and graph schema
- Evidence pointers retained for traceability
- Rapid updates in measures, constraints, and applicability envelopes
- Extension-ready schema to incorporate new categories (tariffs, supply chain finance, sustainability)

A structural advantage of the knowledge graph approach is compounding: each incident processed expands the space of decisions the platform can support, and the matching accuracy improves as the case base grows.

Graph Intelligence and Graph-RAG Agents

ECC's knowledge is not stored as disconnected documents. Graph representation unlocks higher-precision retrieval, compositional reasoning, and explainability.

Why Graph Representation Matters for Resilience

- **Traceability:** parts → materials → suppliers → plants → orders → impact
- **Propagation:** disruptions ripple across dependent nodes and edges
- **Compositional reasoning:** combine measures across categories (inventory + alternates + logistics)
- **Explainability:** every suggestion is backed by incident evidence and relationship paths
- **Queryability:** targeted questions yield precise answers

Graph-RAG agents combine graph traversal (high precision) with large language model synthesis (high usability). This enables a guided copilot that can answer "why" and "what next" questions grounded in both the Knowledge Graph and the customer's Operations Graph — with full traceability back to the specific evidence and constraint that produced each recommendation.

Breadth of Research Categories

ECC's resilience intelligence spans high-impact, decision-relevant genres across nine categories:

- Raw material shortages (rare earth magnets, semiconductors)
- Supplier failures (capacity, quality lapses, insolvency)
- Logistics disruptions (ports, lanes, geopolitical constraints)
- Tariffs and trade disruptions (policy shocks and compliance constraints)
- Sanctions, export controls, and geopolitical policy changes
- Supply chain finance (working capital levers, DPO/DSO impacts)
- Sustainability disruptions (Scope 3, material compliance, eco-design constraints)
- Financial strategies for resilience (risk-to-EBIT, buffer economics)
- Cyber and IT dependencies affecting physical supply chain operations

This breadth powers richer what-if scenarios and unlocks cross-category strategies — for example, a combined tariff shock and material shortage scenario with an emissions compliance overlay — that narrow tools cannot support.

From ICA to PCA: Short-Term Recovery and Long-Term Strategy

Most platforms focus on immediate remediation. ECC is designed to cover both horizons:

- **ICA (Immediate Corrective Actions):** recover production and deliveries as quickly as possible
- **PCA (Permanent Corrective Actions):** eliminate repeat failure modes with structural changes

The Case for Permanent Corrective Actions

PCA requires pattern mining across incidents: identifying what failed repeatedly, why the organisation was exposed, and which structural levers prevent recurrence. This is typically neglected because it demands deep knowledge synthesis across time — but it is where the

largest financial wins often reside.

ICA saves the quarter. PCA saves the company from paying the same disruption tax every year.

Extending into Sustainability and Optimisation

ECC's architecture naturally extends into supply chain sustainability optimisation. This is not just reporting — it is decision-making: choosing levers that reduce emissions while preserving service levels and cost constraints.

The sustainability knowledge layer includes material substitution strategies (engineering feasibility and supplier feasibility), process and routing changes (energy intensity, yield, compliance), low-carbon sourcing options with constraints and lead-time tradeoffs, inventory and buffering strategies that reduce waste and expedite costs, and multi-objective optimisation levers across cost, risk, emissions, and service level.

What-If Scenarios Powered by RIF

What-if planning is only as good as the diversity of measures and constraints that can be simulated. Because RIF spans multiple disruption genres (tariffs, finance, sustainability, geopolitics), it unlocks richer, more realistic what-if scenarios.

Scenario libraries are grounded in real incident archetypes and evidence. Measures are constraint-aware (CACM) rather than hypothetical. Optimisation packaging converts scenarios into executable plans. Cross-category scenarios are possible — for example, tariff shock combined with material shortage combined with emissions cap. Teams collaborate in guided flows rather than static dashboards.

Where Monitoring Platforms Stop and Evidence-Based Optimisation Begins

Incumbent SCRM platforms are strong at mapping and monitoring — detecting events, tracing supplier exposure, and raising alerts at scale. The capability they were not designed for is converting those alerts into ranked, feasible, evidence-backed response plans under live operational constraints.

The architecture described in this paper — two connected graphs, a constraint-aware matching layer, and a mathematical optimisation solver — addresses the capability layer that begins where monitoring ends:

- **Action recommendations:** evidence-grounded and constraint-filtered, rather than generic playbooks
- **Optimisation packaging:** solver-backed plans across supply and production capacity, rather than human-assembled options
- **Long-horizon strategy (PCA):** pattern-mined structural levers, not just immediate firefighting
- **Cross-genre intelligence:** unified knowledge synthesis across tariffs, logistics, geopolitics, and sustainability, rather than category-specific tools

Capability Overview

Capability Layer	Status

Research Knowledge Graph + Operations Graph + CACM recommendations + solver plan packaging	Available
PCA mining, what-if engine, research access agent	Next release
Advanced PCA optimisation, agentic execution	Mid-term roadmap

Glossary

PMG — Product Material Genealogy. Structured mapping from raw materials through components and assemblies to finished products.

Operations Graph — Customer-specific digital twin graph for feasibility and constraint reasoning.

Resilience Research KG — Global incident, measure, and evidence graph maintained by ECC.

RIF™ — Resilience Intelligence Fabric. The intelligence layer that binds the two graphs to drive decisioning.

CACM™ — Constraint-Aware Context Matching. Scenario-to-evidence matching under feasibility constraints.

ICA — Immediate Corrective Actions. Short-term recovery measures.

PCA — Permanent Corrective Actions. Long-term resilience strategies.